

Advanced Data Analysis with R



Welcome

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We are always happy to supervise student projects and theses with a modeling and coding focus!



Classes that might interest you

- [Agent-based modelling with NetLogo](#) (700268, beginners at any level, SoSe, block in late summer)
- Introduction to Ecological Modelling (700549, MSc Forest and Ecosystem Sciences, WiSe)
- Ecological Simulation Modelling (700562 + 700563, MSc Forest and Ecosystem Sciences, SoSe)
- [Spatial Statistics](#) (700294, MSc Forest and Ecosystem Sciences, PhD, SoSe)
- Ecological Modelling with C++ (700300, MSc, WiSe)



Today



Today's Plan

- 🖐️ Welcome (10 min)
- 🌐 Reproducibility & Research Compendia (15 min)
- 💡 Setting Up RStudio for Productive Coding (10 min)
- ⚙️ How to Setup a Research Compendium (live coding, 15 min)
- 💻 Create your own Research Compendium I (Setup, 20 min)
- ☕ Break (10 min)
- ⚙️ Example script and howto turn in a RC (15 min, walkthrough)
- 💻 Create your own Research Compendium II (Code + Data, 30 min)
- 🧩 Why Functions improve your code (15 min incl. live coding)
- 💻 Create your own Research Compendium III (Functions, 25 min)
- ✅ Wrap-up and Q&A (15 min)

Reproducibility & Research Compendia



Reproducibility - Why should we care?

- rerun analyses later
- update code and data safely
- collaborate more easily
- avoid fragile workflows



Reproducibility - Different Levels

1. It runs once in my R console
2. It reruns from a fresh R session (R scripts)
3. Another person can see what to run (understandable code)
4. Another person can reproduce the outputs (code + data + metadata)
5. The workflow is modular, documented, and automated

 **The level depends on the complexity of your project**



Research Compendia

A research compendium bundles:

- data
- code
- outputs
- metadata
- documentation

Recommendation: [The Turing Way](#)



Elements of a Compendium

my-project/

- |— data/
- |— R/
- |— analysis/
- |— output/
- |— README.md
- |— my-project.Rproj

What goes where?

- `data/` → raw and processed data
- `R/` → reusable functions
- `analysis/` → scripts / reports
- `output/` → figures and tables
- `README.md` → overview and instructions

How to Setup a Research Compendium

Setting Up RStudio for Productive Coding

- Go to **Tools > Global Options > Code > Diagnostics**:
 - Enable code diagnostics for better code quality
- Use **soft wrap** and **native pipe operator** (Editing tab), and **rainbow parentheses** (Display tab)
- Have a look around what else you can customize



Create your own Research Compendium I

Setup the basic structure, use RStudio Project

Use a RStudio Project

```
my-project/  
├─ data/  
├─ R/  
├─ analysis/  
├─ output/  
├─ README.md  
└─ my-project.Rproj
```

Then,  Break until XXX

From script to research compendium



Create your own Research Compendium II

Turn Script into a Research Compendium

- one analysis → one file in `analysis/`
- input data → `data/`
- reusable code → `R/`
- outputs → `output/`
- add a `README.md`

Why Functions improve your code (15 min incl. live coding)



Functions

- Functions are reusable blocks of code
- They take inputs (arguments) and return outputs
- Use functions to encapsulate logic and improve code readability

```
1 square <- function(x) {  
2   return(x^2)  
3 }  
4 square(5)
```



Why functions matter

- avoid repetition
- make intent explicit
- reuse logic safely
- hide complexity and simplify scripts



Why functions matter

```
1 bmi_data <- read_csv("data/bmi_data")
2
3
4 #-----
5 # t-Test
6 #-----
7
8 ## Einstichproben-t-Test (One Sample t-Test)
9
10 groesse175 <- rnorm(999, mean=1.75, sd=1)
11 mean(groesse175)
12
13 par(mfrow=c(1, 2))
14 breaks <- seq(from=1, to=2.5, by=0.5)
15 hist(groesse175, breaks=breaks, col="darkblue", las=1,
16      main=paste("p-Wert = ", round(pvalue, 2)))
17 abline(v=1.75, col="darkred", lwd=3)
18
```

```
1
2 data <- read_data() # reads and preprocesses data
3 model_best <- data |> model_select()
4
5 data |> create_fig_1()
6 data |> create_table_1()
7 model_best |> create_fig_2()
8
9 ...
```



Create your own Research Compendium III

Tasks

- simplify your analysis files with functions



Wrap-up and Q&A (15 min)



Wrap-up

- structure matters
- small functions reduce duplication
- reproducibility is built step by step
- a good analysis is also a good workflow



Enjoy the sunny day!



Learning Git

[Learn git concepts, not commands](#): Blogpost that explains really well the concepts of git, also more advanced ones like [rebase](#) or [cherry-pick](#).

[How to write good commit messages](#): Blogpost that explains why good commit messages are important and gives 7 rules for writing them.

[Git cheat sheet](#): Always handy if you don't remember the basic commands

[Book on how to use Git with R](#)

- <https://github.com/selinaZitrone/YoMos2020>
- <https://github.com/baumer-lab/fertile>